

Interaction Context Often Increases Sycophancy in LLMs

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Abstract

We investigate how the presence and type of interaction context shapes sycophancy in LLMs. While real-world interactions allow models to mirror a user’s values, preferences, and self-image, prior work often studies sycophancy in zero-shot settings devoid of context. Using two weeks of interaction context from 38 users, we evaluate two forms of sycophancy: (1) *agreement sycophancy* – the tendency of models to produce overly affirmative responses, and (2) *perspective sycophancy* – the extent to which models reflect a user’s viewpoint. Agreement sycophancy tends to increase with the *presence* of user context, though model behavior varies based on the context *type*. User memory profiles are associated with the largest increases in agreement sycophancy (e.g. +45% for Gemini 2.5 Pro), and some models become more sycophantic even with non-user synthetic contexts (e.g. +15% for Llama 4 Scout). Perspective sycophancy increases only when models can accurately infer user viewpoints from interaction context. Overall, context shapes sycophancy in heterogeneous ways, underscoring the need for evaluations grounded in real-world interactions and raising questions for system design around alignment, memory, and personalization.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

sycophancy, long-context, personalization, mirroring, alignment, memory, human-LLM interaction

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1 Introduction

Sycophancy refers to a broad class of mirroring behaviors in interactions where one party reflects the other’s perspective, values, or self-image [7, 19, 22, 41]. In human interactions, people may exhibit sycophancy to gain approval, persuade others, or foster connection. Some forms of sycophancy are overt ingratiating behaviors, such as offering excessive compliments or showing enthusiastic agreement. Other forms are more subtle, such as downplaying disagreement, adopting the other person’s perspective, or subconsciously mirroring conversation styles. These behaviors arise differently across interpersonal dynamics, suggesting that sycophancy is shaped by and may vary with interaction contexts.

Several recent works show that large language models (LLMs) exhibit sycophantic behaviors, yet these evaluations are often limited to zero-shot settings without user context. One common method of evaluating sycophancy is using rebuttals, such as “Are you sure?”, and measuring whether the model changes its answer [12, 23, 26, 41]. Other studies find that models readily agree with stated claims, even when those claims are subjective or factually incorrect [12, 41]. However, users do not always express their opinions or beliefs explicitly in real interactions. Prior human-computer interaction (HCI) research (§ 2.1) has identified broader forms of AI mirroring behaviors – such as confirmation bias [42], linguistic style matching [30], and emotional contagion [21] – which are often shaped by system design choices around personalization and alignment [11, 43, 52]. Still, evaluations of LLM sycophancy remain limited in *long-context* and *real-world* user interactions (§ 2.2).

Recent frontier LLMs support context windows exceeding one million tokens, enough to include long conversation histories alongside rich digital footprints such as web search or social media activity [8, 31]. Moreover, LLM-based chatbots often have memory features designed to distill user context into salient details that can enhance personalization. While these advances enable contextually rich interactions, they also blur the boundary between personalization and sycophancy, potentially fostering echo chambers and enabling delusional thinking. Over a 300-hour conversation, one ChatGPT user became convinced he had discovered a novel mathematical formula and that he was a real-life superhero [17]. In another case, ChatGPT told a psychiatric patient he could jump off a 19-story building and fly if he believed hard enough [16]. Although these represent extreme cases where sycophancy may have impacted users, they motivate the need to understand how the presence and types of user context shape sycophancy in LLMs.

In this work, we study how user context shapes sycophantic behavior in LLMs using two weeks of real interaction data from 38 participants. Each participant interacted with GPT 4.1 Mini in a persistent context window, yielding an average of 90 queries and 34,416 tokens of context. We use each participant’s conversation history to generate new LLM responses for two tasks: personal advice and political explanations. In personal advice, we evaluate five LLMs for *agreement sycophancy* – advice that is overly agreeable or flattering – using an LLM-judge approach adapted from a prior zero-shot evaluation [7] (§ 3.3). For political explanations, we evaluate two LLMs for *perspective sycophancy*, or the extent to which explanations reflect a user’s political views. To measure perspective sycophancy, participants rated responses generated with and without their context using a 4-point Likert scale (§ 3.4).

We find that agreement sycophancy tends to significantly increase ($p < 0.05$) with the *presence* of user context, though model behavior varies based on the context *type* (§ 4.1). With zero-shot responses as a baseline, we compare agreement sycophancy across responses generated with synthetic interactions, user interactions, and user memory profiles. For Gemini 2.5 Pro, Claude Sonnet 4, and GPT 4.1 Mini, user memory profiles are associated with the largest increases in agreement sycophancy: +45%, +33%, and +16%, respectively. For Llama 4 Scout, user interaction contexts are associated with a 25% increase, while memory profiles do not involve a significant change. GPT 5.1 does not exhibit a significant change with user interactions or memory profiles. Furthermore, some models become more agreeable even in non-user synthetic interactions, such as Llama 4 Scout (+15%) and Gemini 2.5 Pro (+9%).

Perspective sycophancy only rises in interaction contexts where models can accurately infer user perspectives (§ 4.2). Due to constraints in our post-interaction survey, we evaluate perspective sycophancy only for Claude 4 Sonnet and GPT 4.1 Mini. Participants rated how accurately each model understood their political views, based on inferences generated from their interaction context. Claude 4 Sonnet demonstrated a somewhat accurate understanding for 45% of participants, whereas GPT 4.1 Mini did so for 71%. When models accurately inferred users’ political views, perspective sycophancy increased by roughly 0.25 to 0.5 on our 4-point Likert scale compared to contexts where models could not infer user views. Moreover, participants rated that political explanations with and without interaction context reflected their views differently about half the time.

Our study has several implications for the evaluation and design of human-AI interaction systems (§ 5). First, we highlight that evaluations must move beyond static benchmarks and single-turn prompts, as interaction context shapes different forms of sycophancy in heterogeneous ways. Evaluations devoid of context may significantly underestimate the risks of sycophancy in real-world interactions. Second, we discuss how some alignment and personalization methods may promote mirroring of users, raising the question: how can systems personalize without amplifying sycophancy? Finally, we suggest that models may be able to detect when they are mirroring on their own, creating opportunities to design interventions that reduce sycophancy in tasks where it is undesired.

2 Background

We frame LLM sycophancy as a subset of mirroring behaviors within human-AI interaction. We focus on two forms of sycophantic behavior and review prior evaluations of each. Our central contribution is to examine whether *long-context* and *real-world* interactions amplify sycophancy, suggesting that it may be an interaction-dependent mirroring behavior rather than a fixed model property.

2.1 Mirroring in Human-AI Interaction

The fields of psychology and philosophy have long examined how humans mirror or adapt to one another in conversation [4, 32, 40]. Mirroring broadly occurs when one entity reflects the features of another, creating the impression of a copy or representation [9]. People have been observed to mirror one another in body language, speech, appearance, values, desires, and fears [34]. More recently, this phenomenon has been studied in the context of human-AI interaction [21, 30, 33, 42, 46]. For example, Stoeva et al. [46] review body movement mirroring in human-robot interaction, while Morris and Brubaker [33] introduce the notion of “generative ghosts”, or AI systems designed to mirror individuals after death. Of particular relevance to this work is the HCI literature exploring how LLMs mirror individual users.

First, we discuss how LLM mirroring may arise from certain forms of *personalization* or *alignment*. Several works argue for designing LLMs that align to individual users’ values [11, 43, 52]. For example, Fan et al. [11] propose a method for user-driven value alignment, in which users actively guide LLMs to better reflect their values. McIlroy-Young et al. [30] discuss how “mimetic models” can act as a force multiplier for productivity, using the example of email automation in which LLMs mirror a user’s writing style to send messages on their behalf. Sun and Wang [47] further indicate how mirroring may increase user trust and engagement, finding that when models were less friendly, agreeable behavior increased user trust. However, it is important to note that not all forms of personalization involve mirroring [15, 43, 53]. For instance, Shen et al. [43] propose a bidirectional approach to human-AI alignment that aims to avoid the one-way mirroring of LLM responses to human preferences.

Although mirroring represents a form of personalization, prior work also highlights its risks, particularly around creating echo chambers and reducing the diversity in user experiences. Simmons [44] define “moral mimicry” as the reproduction of moral foundational biases associated with user demographics. Sharma et al. [42] find that users engage in more selective search behaviors when using LLMs for political explanations, describing this as a “generative echo chamber”. Jones et al. [21] fine-tune LLMs on individuals’ social media data¹, and discuss how this produces a “mimicry of emotional connection” with users. Many participants in their study described these personalized LLMs as creepy, invasive, and uncanny. Peters and Matz [37] further suggest that mirroring can arise without fine-tuning, demonstrating that LLMs are able to infer users’ personality traits from just 15-turn conversations.

¹While Jones et al. [21] only fine-tune LLMs on social media data as a demonstration, commercial AI assistants are already personalized in this way. Google’s Gemini can incorporate a user’s entire web search history as context [8], while the Meta AI app tailors responses using interaction data from Facebook and Instagram [31].

While the HCI literature has discussed both the benefits and risks of AI mirroring, open questions remain about how and when such behaviors emerge in long-context, naturalistic LLM interactions. There also remains a general evaluation gap of how model behavior changes in long-context settings. Most previous evaluations of LLM behavior in long-context rely on synthetic data [2, 27], citing the lack of publicly available collections of extended interactions. For example, in WildChat, a commonly-used dataset for real-world user queries, the average interaction length is only 2.5 conversation turns per user [54]. Moreover, previous evaluations focus on benchmarking LLM performance or recall in long-context, not on evaluating how user contexts shape model behavior.

2.2 Evaluations of LLM Sycophancy

Sycophancy in LLMs broadly refers to excessive mirroring behaviors in which models are overly agreeable with or reflective of users [7, 41]. In this work, we focus² on two forms of LLM sycophancy, which we define below before reviewing prior evaluations of these behaviors.

Definition 2.1 (Agreement Sycophancy). Model behavior that excessively mirrors a user’s positive self-image through overly agreeable or flattering responses.

Definition 2.2 (Perspective Sycophancy). Model behavior that excessively mirrors a user’s perspective or viewpoint in responses.

Note that these definitions are not mutually exclusive; a model may exhibit agreement sycophancy, perspective sycophancy, or both.

Agreement Sycophancy. A common evaluation method for agreement sycophancy is using *rebuttals*, such as “Are you sure?”, and measuring whether the model changes its answer [12, 18, 23, 26, 41]. Sharma et al. [41] show that LLMs often change their answers to factual questions when prompted with such rebuttals, even after initially responding correctly. Fanous et al. [12] distinguish between preemptive rebuttals, where the user’s stance is included in the original prompt, and in-context rebuttals, where the rebuttal is provided in a second turn after the original response. They find that preemptive rebuttals yield agreement more frequently than in-context rebuttals, suggesting that sycophancy is more prevalent when users are upfront and explicit about their beliefs. Wang et al. [50] further show that model agreement is higher when evaluation prompts include first-person beliefs rather than third-person beliefs.

While most evaluations of agreement sycophancy focus on settings with a ground truth, a few works evaluate agreement in open-ended settings such as personal advice and ethical dilemmas [7, 18, 28]. In particular, Cheng et al. [7] evaluate sycophancy using personal advice scenarios from Reddit’s “Am I the Asshole” (AITA) forum. They define *social sycophancy* as the excessive preservation of a user’s “face” (the positive self-image a person seeks to maintain in an interaction). In 42% of cases, models affirm behavior deemed inappropriate by crowd-sourced human judgments on

²Previous works define many model behaviors under the construct of sycophancy. For example, Sharma et al. [41] define sycophancy as model responses that match user beliefs over *truthful* ones. Another form is “indirectness sycophancy”, or model behavior that hedges or provides vague suggestions instead of clear statements [7]. While these are outside the scope of our work, they also remain understudied in long-context and real-world interactions.

Reddit. To measure agreement, they use an LLM-judge to classify whether responses to the advice scenarios indicate any wrongdoing by the user. We adopt Cheng et al. [7]’s evaluation of agreement sycophancy in our work because it captures AI mirroring of the user’s implicit desire for validation and approval, beyond simple susceptibility to rebuttals.

Perspective Sycophancy. We use the term “perspective sycophancy” to describe model behavior that excessively mirrors a user’s perspective or worldview. In contrast to agreement or affirmation behaviors, perspective sycophancy concerns the framing of information. A model may adopt a user’s ideological lens or interpretive stance without explicitly endorsing it. For example, a model may describe a news article from a liberal or conservative perspective that aligns with the user’s political ideology without directly affirming it. When such perspective mirroring becomes “excessive” depends on normative interpretation; accordingly, we view evaluations of perspective sycophancy as closely related to those of personalization and alignment in LLMs.

A common method for evaluating perspective sycophancy is to use *personas* and measure how closely model responses align with the demographics or ideologies those personas represent [6, 24, 44, 49]. For example, Simmons [44] show that models exhibit political bias when prompted with liberal or conservative personas. Across different sociodemographic personas, Kim et al. [24] find that political personas produce the largest shifts in LLM decision-making. In alignment evaluations, users commonly assess pairs of responses to judge which more closely aligns with their values or preferences [25, 43]. Our evaluation of perspective sycophancy follows a similar approach: participants compare responses generated with and without their interaction context and rate how closely each reflects their perspective.

3 Data and Methods

In this section, we first describe our participant pool (§ 3.1) and how we collected two weeks of LLM interaction data for each participant (§ 3.2). We then describe our evaluations of agreement sycophancy in personal advice (§ 3.3) and perspective sycophancy in political explanations (§ 3.4). In each evaluation, we compare LLM responses generated with and without each participant’s interaction context. We draw on an LLM-judge approach from prior work [7] to measure agreement sycophancy, and use participant ratings from a post-interaction survey to measure perspective sycophancy. We use a regression analysis to study how each form of sycophancy relates to the presence and type of context (§ 3.6).

Table 1: Number of Participants by Gender & Political Views

	Very Liberal	Liberal	Moderate	Conservative	Very Conservative	Total
Men	4	2	6	3	4	19
Women	5	5	3	3	1	17
Non-Binary	1	0	1	0	0	2
Total	10	7	10	6	5	38